Prasanna **Maddila**

Algorithmic Game Theory | Reinforcement Learning

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Education

University of Toulouse | Lab MIAT, INRAE — Toulouse, France

PhD in Computer Science — Oct 2023 - Oct 2026

- Worked on the Nash Equilibrium computation in cooperative-competitive games.
- Implemented GPU-compatible solvers in JAX for proposed algorithms, and RL environment for application case.
- PhD funded by the French-German ANR Project CHIP-GT.

CentraleSupélec — Paris, France

Masters in Engineering (Computer Science) — August 2021 - 2023

- Specialisation in Theoretical Computer Science, and part of the Research Track (*Filière de Recherche*).
- Student in the Accelerated Engineering Degree Programme.

Mahindra University — Hyderabad, India

Bachelors' in Technology. Mechanical Engineering — August 2017 - 2021

- Recipient of the University's Student Scholarship for Academic Excellence for 3 years.
- Member of the Organising Team TEDx (2018-19), and Vice-President of the Student Council (2020).

Experience

End-of-Studies Internship — Toulouse, France

MIAT, INRAE — Apr - Sept 2023

- Modelled Anti-Poaching as a Partially Observable Markov Game.
- Implemented a simulator using PettingZoo, with various MARL algorithms using RLlib and PyTorch.

M2 Project — Paris, France

MICS, CentraleSupélec — Nov 2022 - Apr 2023

- Worked on the Roommate Matching problem under Rank-Envy-Free matching constraints.
- Established complexity results for such matchings, and studied their relation to existing notions.

M1 Internship — Paris, France

MICS, CentraleSupélec — May - July 2022

- Worked on Epidemiological models to model the evolution of COVID Variants.
- Implemented numerical schemes and calibrated models using real-world data.

Visiting Student — Shantou, China

University of Shantou — May - July 2018

- Developed approaches based on genetic algorithms for modelling of Li-ion batteries in Electric Vehicles.
- Worked on data analysis and Experimental Design to minimise physical testing.

Teaching

Deep Learning — Toulouse, France

University of Toulouse — November 2024 - December 2025

- Created and taught Deep Learning using PyTorch to M2 Mathematical Modelling of Ecological Systems students.
- Course projects: Auto-Encoders for Image Segmentation and training CNN Classifiers for real-world datasets.

Skills

Programming Languages	Python, C, C++17, MATLAB, HTML/CSS (Basics)
Libraries	JAX, PyTorch, RLlib, NumPy, Pandas, Matplotlib, Pytest, Sphinx, WandB
Other Software/Skills	LaTeX, Make, CMake, Bash, Git, CI/CD, Docker (Basics)
Spoken Languages	English (C1/C2), French (B2), Hindi, Telugu

Software

Anti-Poaching Environment

MIAT, INRAE

- Reference implementation of the Anti-Poaching Environment, implemented using PettingZoo.
- Implemented MARL algorithms using RLlib/BenchMARL to leverage multiple CPUs or GPU using PyTorch backend.
- Simulator implements a formally prescribed Partially Observable Markov Game (POMG) model.

Multi-Adversarial Team Games

MIAT, INRAE

- JAX-based, GPU-compatible implementation of the solver for Multi-Adversarial Team Games.
- Implementation scales to 4v9 games with 8 actions each, and can be vectorised to solve multiple games in parallel batches.

Finite-Horizon Multi-Adversarial Team Markov Games

MIAT, INRAE

- JAX-based, GPU-compatible implementation of the solver for Markov extensions of Multi-Adversarial Team Games.
- Vectorised implementation with unit tests and typing annotations for correctness.

Conferences and Workshops

Maddila, P., Sabbadin, R., & Vinyals, M. (2026, May). Efficiently Computing Approximate Nash Equilibria in Multi-Adversarial Team Games. *The 25th International Conference on Autonomous Agents and Multi-Agent Systems*. https://hal.science/view/index/docid/5450199

Maddila, P., Sabbadin, R., & Vinyals, M. (2026, August). Approximating Nash Equilibria in Finite-Horizon Multi-Adversarial Team Markov Games. *Forty-Second Annual Conference on Uncertainty in Artificial Intelligence*. https://openreview.net/forum?id=yXg85NFLkO

Maddila, P., Sabbadin, R., & Vinyals, M. (2024, March). Anti-Poaching as a Partially Observable Stochastic Game. *25ème édition du congrès annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision ROADEF 2024*. https://hal.science/hal-04525926

Maddila, P., Sabbadin, R., & Vinyals, M. (2024, July). Learning in Competitive-Cooperative Games for Anti-Poaching. *Workshop on Learning in Games*.

Maddila, P., Casellas, E., Chabrier, P., Sabbadin, R., & Vinyals, M. (2024, October). APE: An Anti-poaching Multi-Agent Reinforcement Learning Benchmark. *Seventeenth European Workshop on Reinforcement Learning*. https://openreview.net/forum?id=JSWRnHC93W

Maddila, P., Sabbadin, R., & Vinyals, M. (2024, November). Un modèle de jeu stochastique pour la lutte anti-braconnage et un benchmark pour l'Apprentissage par Renforcement Multi-agents. *JFSMA 2024: 32èmes Journées Francophones sur les Systèmes Multi-Agents*. https://hal.science/hal-04699116

Coutance, B., Maddila, P., & Wilczynski, A. (2023, August). Rank-Envy-Freeness in roommate matchings. *26th European Conference on Artificial Intelligence (ECAI 2023)*.

Agrawal, A., Maddila, P., Zhang, C., Perabathini, B., & Lasaulce, S. (2019). Cooperative energy efficient resource allocation in fast fading interference networks. *2019 International Conference on Wireless and Mobile Computing, Networking and Communications (Wimob)*, 276–281.

Bulusu, S. K. C., Maddila, P., Shaiek, H., & Roviras, D. (2019). HPA linearization for FBMC-OQAM signals with fast convergence-digital predistortion. *2019 16th International Symposium on Wireless Communication Systems (ISWCS)*, 133–138.

Journal Publications

Yun, L., Li, W., Garg, A., Maddila, P., Gao, L., Fan, Z., Buragohain, P., & Wang, C.-T. (2019). Maximization of extraction of Cadmium and Zinc during recycling of spent battery mix: An application of combined genetic programming and simulated annealing approach. *Journal of Cleaner Production*, *218*, 130–140.

Yun, L., Maddila, P., Gao, L., Peng, X., Niu, X., Garg, A., & Chin, C. M. M. (2019). An integrated framework for minimization of inter lithium-ion cell temperature differences and the total volume of the cell of battery pack for electric vehicles. *Energy Storage*, *1*(2), e41.